

Application machine learning for risk prediction using synthetic lung cancer patient data

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Abstract—Lung cancer stands as one of the primary causes of cancer-related fatalities on a global scale. Machine learning applications have been an ideal approach in predicting patients' symptoms and medical records. The primary aim of this. The primary aim of this research is to evaluate 3 machine learning models, extreme gradient boost (XGBoost), support vector machine (SVM), and logistic regression (LogReg) to yield the best results in predicting whether a patient having lung cancer based on their symptoms and medical records. MITRE's open-source Synthea platform was used to train these machine learning models on the Jupyter notebook using Python 3 (ipykernel); with its given dataset that has thousands of synthetic patients and 791 features that contribute to identifying lung cancer. The machine learning models are from the sklearn, and xgboost libraries that are used for training with the dataset. After the preprocessing, data exploration, baseline modeling and hyperparameter tuning stages, the results show that the XGBoost model had a higher recall value than the two models. A higher recall value indicates, based on the confusion matrix, a lower count of false negatives, which is crucial to identifying a lower rate of patients succumbing to lung cancer. XGBoost is a highly recommended model for medical diagnosis use cases.

Keywords—machine learning, lung cancer, support vector machine, extreme gradient boost, logistic regression, grid search, random search, hyperparameter tuning, normalization, features, labels, instances, dataset, pipeline

I. INTRODUCTION

A. Review of Related Literature

Lung cancer stands as a formidable global health challenge, with an alarming annual mortality rate of approximately five million cases, surpassing even the combined mortality rates of breast and prostate cancer [14]. Despite its severity, early detection and diagnosis are crucial for improving survival rates [14]. While various modalities are employed for detecting and diagnosing lung cancer, Computed Tomography (CT) scans are particularly effective, providing significant information regarding lung infections [14]. The advent of advanced imaging and sequencing technologies has led to an explosion of complex data in lung cancer research, which can overwhelm human analytical capabilities [15]. This is where machine learning (ML) approaches become indispensable [15].

ML-based techniques are revolutionizing the field by integrating and analyzing these vast datasets, offering new perspectives on lung cancer [15]. Researchers have developed novel ML techniques for the detection and classification of

different types of lung cancers, including Adenocarcinoma, Large cell carcinoma, and Squamous cell carcinoma [14]. These techniques often involve feature extraction and fusion using methods like patch-based Local Binary Pattern (LBP) and Discrete Cosine Transform (DCT) [14]. Machine learning algorithms such as Support Vector Machine (SVM) and K-Nearest Neighbors (KNN) are then used to classify texture features from CT scan images [14]. For instance, one study demonstrated that these techniques achieved accuracies of 93% with SVM and 91% with KNN for lung cancer detection, outperforming existing state-of-the-art methods [14].

Lung cancer risk prediction demands advanced analytical approaches due to its complexity, high mortality, and diverse contributing factors (e.g., genetics, imaging, environmental exposures) [1]. Traditional statistical models often fail to capture nonlinear, high-dimensional relationships across large datasets [5]. ML is a transformative tool due to its ability to integrate multi-modal data, enable personalized medicine, and offer early detection with high accuracy [2]. ML can identify subtle patterns from radiomics, genomics, and clinical data, enhancing predictive performance and guiding individualized interventions. The process of applying ML to lung cancer risk prediction involves three major stages: model building, evaluation, and improvement [10]. During model building, relevant patient data are collected, cleaned, and processed, this includes tasks such as normalizing values, encoding categorical variables, and selecting predictive features like pack-years of smoking or imaging-derived tumor characteristics [11]. The data is then used to train models that deal in classification of lung cancer such as logistic regression, support vector machine, and extreme gradient boost, which learn to map input variables to predicted lung cancer risk scores [7][8][13]. Evaluation follows, using performance metrics such as precision, recall, and F1-score to assess how well the model performs in identifying high-risk individuals, with particular emphasis on minimizing false negatives to avoid missing true cases. Cross-validation techniques ensure robustness and generalizability [9]. The models are improved through iterative refinement, involving hyperparameter tuning, and the grid search method to boost predictive power [6]. After evaluating the three models' performance after improvements, the best model is picked for deployment in practical use in risk prediction for lung cancer.

Other than just detection and classification, ML is playing a pivotal role in various aspects of lung cancer management, including early detection, auxiliary diagnosis, prognosis

prediction, and optimizing immunotherapy practices [15]. As the field progresses, the application of machine learning in lung cancer diagnosis, treatment, and prognosis presents both exciting opportunities and unique challenges for future advancements [15].

B. Dataset Description

(MITRE)'s Synthea platform creates realistic synthetic health records, produced the dataset used in this investigation [1]. Machine learning-ready tables centered on lung cancer cases and controlled patient samples were created from five cohorts, each containing 30,000 synthetic individuals. The dataset is available on Harvard Dataverse and has been cited in a published publication, helping machine learning research in learning health systems [1]. The dataset's goal is to create and test machine learning models for lung cancer risk prediction, serving as a privacy-preserving substitute for real patient data and simulating a learning health system workflow to improve early lung cancer detection and intervention in real clinical settings [1].

Each row represents each instance that includes the patient's assigned unique number used for anonymity and the values that correspond to each Systematized Nomenclature of Medicine—Clinical Terms (SNOMED-CT) feature; each column represents features that are relevant to the patient. They are coded and catalogued as SNOMED-CT. These columns that has over 791 represents features that are categorized by the following:

- **Demographics and Social Determinants**
 - **Basic Demographics:** race, gender, age, marital status, ethnicity, and state
 - **Social Factors:** employment status, homelessness, refugee status, and social migrant status
- **Vital Signs and Basic Measurements**
 - **Core Vitals:** blood pressure, heart rate, respiratory rate, body temperature, and oxygen saturation
 - **Anthropometric Measures:** Body weight, height, Body Mass Index (BMI)
- **Laboratory Tests and Biomarkers**
 - **Lipid Panel:** total cholesterol, high-density lipoprotein (HDL), low-density lipoprotein (LDL), and triglycerides
 - **Complete Blood Count (CBC):** Hemoglobin, hematocrit, red blood cell count (RBC), white blood cell count (WBC), platelet count, mean corpuscular hemoglobin (MCH), mean corpuscular hemoglobin concentration (MCHC), mean corpuscular volume (MCV), red cell distribution width (RDW), platelet distribution width
 - **Basic Metabolic Panel (BMP):** Glucose, sodium, potassium, chloride, carbon dioxide (CO₂), creatinine, blood urea nitrogen (BUN), calcium
- **Liver Function Tests (LFTs):** Alanine transaminase (ALT), aspartate transaminase (AST), bilirubin, albumin, alkaline phosphatase, total protein, globulin
- **Cardiac Markers:** Troponin, N-terminal pro b-type natriuretic peptide (NT-proBNP), lactate dehydrogenase
- **Specialized Tests:** Hemoglobin A1c (HbA1c), Prostate-Specific Antigen (PSA), urine microalbumin, glomerular filtration rate (GFR), procalcitonin, D-dimer, C-reactive protein (CRP), ferritin, iron studies
- **Coagulation Tests:** Prothrombin Time (PT), International Normalized Ratio (INR)
- **Hormones:** Thyroid Stimulating Hormone (TSH), free thyroxine (free T4)
- **Urinalysis:** Multiple urine tests including protein, glucose, ketones, and microscopic analysis
- **Screening and Assessment Procedures**
 - **Mental Health Assessments:** Patient Health Questionnaire-2 (PHQ-2), Patient Health Questionnaire-9 (PHQ-9), anxiety assessment, substance use assessment, Alcohol Use Disorders Identification Test - Consumption (AUDIT-C), suicide risk assessment
 - **Preventive Care:** Colonoscopy, mammography, bone density scan, fecal occult blood testing, drug abuse screening
 - **Specialized Screenings:** Domestic abuse screening, fall risk assessment (Morse Fall Scale), health and social care needs assessments
- **Chronic Diseases and Conditions**
 - **Cardiovascular Diseases:** Hypertension, coronary heart disease, atrial fibrillation, heart failure, myocardial infarction (heart attack), stroke
 - **Metabolic Conditions:** Diabetes, prediabetes, metabolic syndrome, hyperlipidemia, hyperglycemia
 - **Neurological Disorders:** Alzheimer's disease, epilepsy, seizure disorders, traumatic brain injury (TBI), concussion
 - **Musculoskeletal Conditions:** Osteoporosis, osteoarthritis, rheumatoid arthritis, fibromyalgia, chronic pain
 - **Respiratory Diseases:** Asthma, Chronic Obstructive Pulmonary Disease (COPD), pulmonary emphysema, pneumonia, Acute Respiratory Distress Syndrome (ARDS)
 - **Gastrointestinal Conditions:** Cholecystitis, appendicitis, polyps, inflammatory bowel conditions

- **Genitourinary Disorders:** Chronic kidney disease (CKD, stages 1–3), urinary tract infection (UTI), cystitis, pyelonephritis
- **Mental Health Disorders:** Depression, anxiety, Post-Traumatic Stress Disorder (PTSD), stress-related disorders
- **Infectious Diseases:** Coronavirus Disease 2019 (COVID-19), influenza, sinusitis, pneumonia, sepsis
- **Oncological Conditions:** Various cancers (including prostate, breast, and colon) with associated cancer staging
- **Acute Conditions and Injuries**
 - **Trauma and Injuries:** Fractures (ankle, wrist, forearm, hip, spine), lacerations, burns, sprains, dislocations
 - **Acute Medical Conditions:** Appendicitis, acute bronchitis, viral pharyngitis, acute sinusitis, cardiac arrest
 - **Medical Complications:** Sepsis, septic shock, acute kidney injury (AKI), Acute Respiratory Distress Syndrome (ARDS)
 - **Obstetric Emergencies:** Miscarriage, ectopic pregnancy, preeclampsia, eclampsia, labor complications
- **Medications**
 - **Cardiovascular Medications:** Angiotensin-Converting Enzyme (ACE) inhibitors (e.g., lisinopril), beta-blockers (e.g., carvedilol), diuretics, statins, antiarrhythmics
 - **Pain Management:** Acetaminophen, Non-Steroidal Anti-Inflammatory Drugs (NSAIDs), opioids (e.g., oxycodone, hydrocodone, fentanyl), combination pain medications
 - **Diabetes Medications:** Insulin, metformin, other antidiabetic agents
 - **Antibiotics:** Penicillin, cephalosporins, macrolides, combination antibiotics
 - **Respiratory Medications:** Bronchodilators, corticosteroids, respiratory therapies
 - **Oncology Treatments:** Chemotherapy agents (e.g., docetaxel, paclitaxel, carboplatin), hormone therapies
 - **Hormonal Therapies:** Contraceptives, hormone replacement therapy, thyroid medications
 - **Psychiatric Medications:** antidepressants, anti-anxiety medications (anxiolytics), mood stabilizers
 - **Specialty Drugs:** Immunosuppressants, anticoagulants
- **Procedures and Interventions**
 - **Diagnostic Procedures:** Imaging studies (X-ray, Magnetic Resonance Imaging [MRI], Computed Tomography [CT]), endoscopy, biopsies, cultures
 - **Surgical Procedures:** Appendectomy, cholecystectomy, cardiac surgeries, orthopedic surgeries, cancer-related surgeries
 - **Therapeutic Interventions:** Physical therapy, occupational therapy, speech therapy, respiratory therapy
 - **Obstetric and Gynecological:** Cesarean section, episiotomy, contraceptive procedures, prenatal care
 - **Cardiac Interventions:** Cardioversion, catheter ablation, coronary artery bypass graft (CABG) surgery
 - **Emergency Interventions:** Endotracheal intubation, mechanical ventilation, cardiopulmonary resuscitation (CPR), emergency medication administration
 - **Specialized Treatments:** Dialysis, organ transplant procedures, radiation therapy, chemotherapy administration
- **Reproductive Health and Obstetrics**
 - **Prenatal Care:** Fetal monitoring, prenatal ultrasounds, prenatal screenings, genetic testing
 - **Pregnancy Complications:** Preeclampsia, gestational diabetes
 - **Labor & Delivery:** Vaginal delivery, cesarean section, labor complications
- **Patient Care and Administrative**
 - **Care Coordination:** Discharge planning, care plan creation, medication reconciliation
 - **Healthcare Utilization:** Hospital admissions, Intensive Care Unit (ICU) transfers, skilled nursing facility stays
 - **Support Services:** Home health care, hospice services, social work services
 - **Documentation:** Electronic medical records, clinical assessments, follow-up care documentation
- **Allergies and Immunology**
 - **Environmental Allergies:** Tree pollen, grass pollen, dust mites, mold, animal dander
 - **Food Allergies:** Peanuts, shellfish, milk, eggs, wheat, soy
 - **Other Allergens:** Latex, bee venom, medication allergies
 - **Allergy Testing:** Immunoglobulin E (IgE) testing for allergen-specific sensitivity
- **Substance Use and Behavioral Health**

- **Substance Use:** Tobacco smoking status, alcohol use disorders, illicit drug use, opioid misuse
- **Behavioral Interventions:** Substance use screening, behavioral therapy, counseling
- **Risk Factors:** Excessive alcohol consumption, tobacco dependence, substance misuse

Depending on the features of the dataset how it classifies can be defined numerically on vital signs and basic measurements features, and medication features, descriptive classification on demographics and social determinants feature binary classification where it indicates a yes or no question on acute conditions and injuries feature, screening and assessment procedure feature and more, or missing value can be present in many of the features indicated as (NaN), since not all instances of patients have recorded data on some of the features.

II. METHODOLOGY

A. Data Preprocessing and Cleaning

The Synthea dataset was gathered under the domain of Harvard's Dataverse [1]. The ZIP file that was downloaded included multiple comma-separated value (.csv) excel files. These contained the various records of patients that were separated into 5 different files. One .csv file also contained the labeling of each feature since each feature in the patient records were assigned in Systematized Nomenclature of Medicine – Clinical Terms (SNOMED-CT) codes. After proper scrutinization of the datasets, the group agreed to use the excel file with a population of 60,000 or named as “synthea-pt60k-lc-data-with-names.csv” because records with higher populations were not optimal and computationally expensive to utilize. Therefore, it makes it easier to do various data cleaning processes, such as one-hot encoding, feature selection, standardization, and others. With that in mind, the 60,000-population dataset was then loaded in either two platforms such as Google Colab or Jupyter Notebook depending on the accessibility of each group member. Because of the unreadability of the dataset since each feature is in SNOMED-CT format, the plan was to map the codes from “synthea-lc-dataset-codes.csv” or the code file to the columns of “synthea-pt60k-lc-data-sel-convert.csv.” A mapping dictionary was incorporated into the map. Each name of the legend from the columns in a mapping dictionary was used to store the codes along with each SNOMED-CT. Using DataFrame from the pandas library, preprocessing strategy is significantly streamlined by utilizing the pre-processed synthea-pt60k-lc-data-sel-convert.tab dataset, a decision justified by an observation from a study that model performance for lung cancer prediction reaches a near-optimal plateau with 60,000 patients, yielding only marginal gains with larger datasets at a substantially higher computational cost [2]. This approach allows to bypass complex and time-consuming consolidation, one-hot encoding, and initial feature selection, enabling a focus on core machine learning tasks like model training, hyperparameter tuning, and in-depth error analysis. The initial steps involve loading clean column names from synthea-lc-dataset-codes.csv and assigning them to the main data file, ensuring the correct (8484, 791) tabular structure. After separating features (X) and target (y), a thorough inspection confirms the necessity of the full preprocessing pipeline due

to a mix of numerical types, categorical strings (e.g., 'normal', 'white'), and significant missing values (NaN). The pipeline addresses these issues through three essential steps: imputing missing values by replacing NaN with -1 to retain information about absence, manually encoding columns with clear semantic meanings (e.g., 'normal'/'abnormal', 'TRUE'/'FALSE') to integers for efficiency and semantic correctness, and finally, one-hot encoding the remaining non-numeric columns like 'race' and 'marital status' into a numerical format. To further refine the dataset, a comprehensive feature selection process is implemented to improve model performance, reduce overfitting, and decrease computational cost. This involves three key stages: ranking features by predictive power using Random Forest's feature importance, finding the optimal number of top features through cross-validation, and removing collinear features to create a final, efficient feature set.

The first step was to parse the column headings in the dataset that are cryptic in nature, e.g., C-103579009 to the human readable name of the column, e.g., race, which was provided by the file, synthea-lc-dataset-codes.csv. This meant that clear clinical context could be used to conduct all analyses and feature selection after that. In the process, some duplicate column names that were produced in the mapping were categorically renamed in order to avoid confusion. Two columns were specifically avoided before any other manipulation was done. A column with the name ptnum was removed since this is a patient identifier column and does not contain any information that may be used in prediction. What is more significant, the scc (Squamous Cell Carcinoma Antigen) column was eliminated. This study aims at developing an early detection model to screen lung cancer, that is, the model must depend on the data that is at hand prior to the existence of a high suspicion of cancer. The SCC is a biomarker that is normally measured once such a suspicion is already created. One of the analyses carried out at the initial stage proved that it has an extreme influence, as the value of its correlation coefficient is about -0.8 with the target variable. The introduction of such a feature would result in immense amounts of target leakage, a kind of "cheat sheet" given to the model which would artificially improve the model performance during the testing version, but fail miserably when it comes to actually screening sufficiently before the established diagnosis.

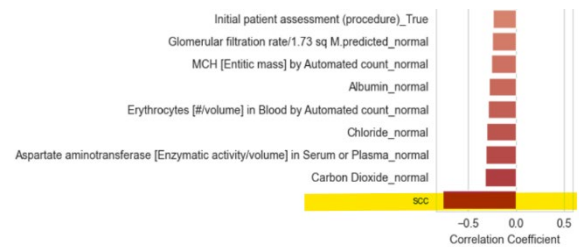


Fig. 1: SCC found to be an overwhelming outlier with an approximately -0.8 correlation coefficient

To make the process of model evaluation methodologically sound, we divided the whole data into training set (80%), and testing set (20%), both stratified on the target variable to have the same distribution of classes in the training and testing set. Such a split-then-preprocess paradigm is fundamental, because it ensures that the training data

determines all the decisions on data cleaning, feature transformation, whereas the test set is a genuine holdout used only at the end to validate the results and avoiding data leakages. The first feature filtering was carried out after the split. Any columns in the training set with more than 60% of its values missing, or displayed zero variance (i.e. had only a single distinct value) were dropped in the training and the test sets. It played an important role in decreasing noise and complexity of the computation through the removal of features that are too sparse and/or uninformative to be of any meaningful contribution to the model.

The second step consisted in a context-specific imputation approach in which imputation rules were made exclusively for the training data: clinical event columns would be filled with a 0 to indicate the absence, categorical columns with a special Unknown category, and all other numerical columns would be imputed using their median value. After imputation, all other categorical variables left in the dataset were numericized through one-hot encoding. Lastly, it should be known that various machine learning models have diverse input needs, thus, two data pipelines were built. In case of models with sensitivity level to feature scale like Logistic Regression and Support Vector Machines, the data after preprocessing data was scaled with standard scaler. At the same time, one of the versions of the data, without scaling, was intentionally kept to be used in the XGBoost model since tree-based algorithms do not depend on the magnitude of features, and their performance is independent of scaling. Such a practice of two parallel pipelines made it possible that every algorithm was trained with data in the most optimal form, therefore resulting in maximum potential performance of each model tested.

B. Feature Selection

We utilized a three-step feature selection protocol after providing the extensive data cleaning operation that provided a pool of 105 possible predictive features. The main aims of that process consisted in reducing the curse of dimensionality, decreasing time of model training, and most importantly, improving the generalization to unseen data by removing irrelevant and redundant features. Initial procedure consisted in ranking of all 105 features in terms of predictiveness. The reason why we selected Random Forest Classifier to perform this task is that the latter inherently computes feature importance scores of Gini impurity. The idea is to evaluate the relevance of each feature in terms of the decreased impurity (or increased purity) it causes in the decision tree nodes in the entire forest, giving a solid, model-based measure of the relevance of each feature with regard to predicting lung cancer. This baseline ranking proved that demographic variables (such as `age_gt70` and `gender_m`) were indeed quite significant, however, it also suggested the promise of certain clinical measures and social determinants of health as predictors.

The latter phase was meant to identify the ideal number of such leading features that were to be maintained empirically. Merely picking a randomly chosen subset of features can also result in performance being less than optimal; either too little data can be lost in the feature selection, or too much information can be kept in the features. To minimize this, we used an iterative cross-validation method. We systematically crossed F1-score of a Random Forest model on consecutively larger subsets of top-ranked features (e.g. top 10, 20, 30, and

so forth). In the process, the results showed that the model worked best with the first 60 features (F1-score = 0.8900). Since this part, further addition of more features would attract an output of diminishing returns and a slight deterioration of performance, a pointer to the addition of noise. This was a data driven method that enabled us to find the optimum point where signal to noise ratio will be maximized. The third and last step dealt with the occurrence of multicollinearity when features are over correlated to one another. Redundant attributes can destabilize some machine learning models and create unneeded complexity. We calculated a correlation matrix of the 60 highest-performing features and we determined and dropped one feature of any pair whose Pearson correlation coefficient exceeded 0.95. In the course of this process 8 redundant features were removed, including the two highly correlated ($\text{correlation} \times 100 = 866$) Initial patient assessment (procedure)_True and Development of individualized plan of care (procedure)_True. The eventual culling left a lean, strong and non-redundant collection of 52 features, halving the starting feature space and empirically improving the base F1-score and placing a high mark on the modeling phase that follows.

C. Exploratory Data Analysis

The scope of the Exploratory Data Analysis covers the target variable distribution, visualizing the missing value, univariate, bivariate, multivariate, and outlier analyses. The target variable distribution involves utilizing the countplot or histograms as seen in Fig. 1.

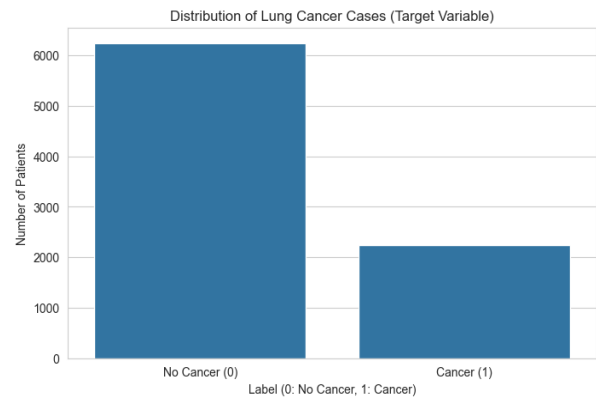


Fig. 2: Target Variable Distribution

Fig. 2, models or maps show how each instance or the number of patients is diagnosed with lung cancer or not. Because the labels are either 1 or 0, each of them is mapped to each patient based on their own instances. Thus, the figure suggests that the 60,000-record dataset contains mostly patients with a less association to lung cancer.

The missing values section showcases the in various cells throughout the Synthea dataset back in the data cleaning stage. This is one of the main reasons why data imputation is highly significant in this pipeline. This is to show the percentage of missing values in each column. The code calculates the frequency and total number of missing values for each column or feature. The code then calculates the percentage of the

missing value per column. The graph shown in *Fig. 2* is a horizontal bar plot from the seaborn library.

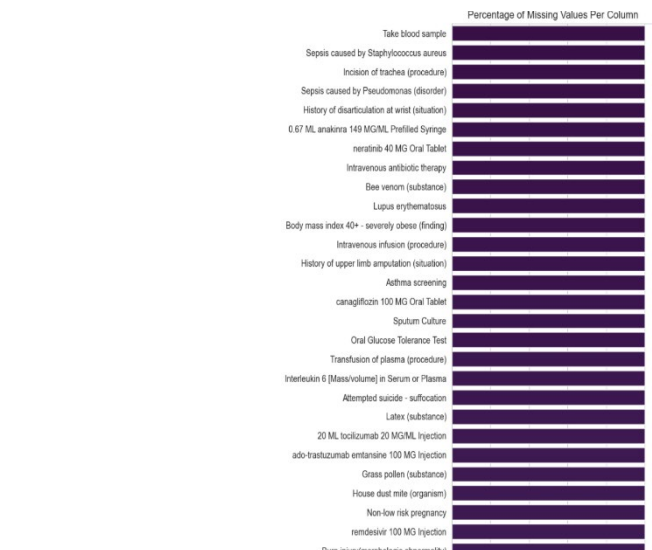


Fig. 3: Upperbound of Missing Values

The graph above shows the percentage of missing values of each column or feature. The highest percentage of missing values is color-coded with a purplish gradient. “Take blood sample,” “Sepsis caused by Staphylococcus aureus,” and “incision of trachea (procedure)” are the three highest features with missing values. The lack of values and observations from these columns would not create a significant impact when feeding it to the model and may cause inaccuracies. Thus, data imputation and other data cleaning methods were used in the process.



Fig. 4: Lower Bound of Missing Values

The bottom of the bar plot includes percentages of all the features that have little to no missing values. This is represented by a yellow-greenish gradient. The three lowest values include “Triglycerides,” “Body Mass Index,” and “Assessment and social care needs (procedure).” These columns are highly important because it would mean that they would not undergo data imputation and would be selected candidates for feature selection.

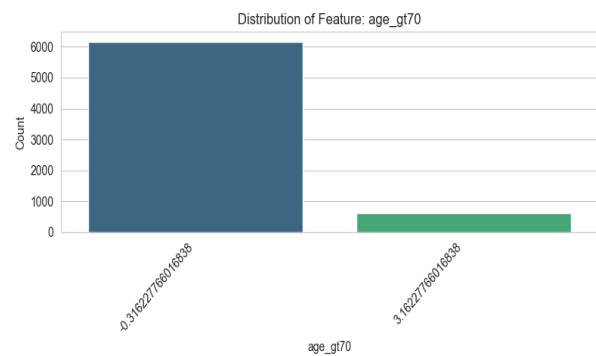


Fig. 5: Univariate Analysis – age_gt70

Univariate analysis highlights the relationship of a single feature to the frequency of observations it has. The coding process basically involves using countplot, while the unique values that are used for the graph were derived through one-hot encoding due to their previously categorical nature. They were then scaled through standardization. In this case, the standardized values become numerical representations. Thus, these two decimal labels would serve if a person is older than 70 or not. Based on the graph, it is apparent that there are a great number of patients who are younger than 70.

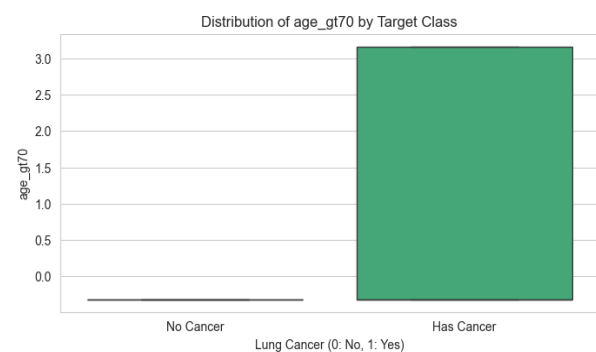


Fig. 6: Bivariate Analysis – age_gt70

Bivariate analysis involves the relationship between two features. The code process used for this is similar; however, the graph used is a box plot, which does not involve bins. This allows for the visualization of how the age_gt70 (the age of patients greater than 70) distribute well with lung cancer detection. Apparently, based on the graph, there are more patients greater than 70 that are associated with lung cancer. The y-axis represents the standardized numerical representation that determines if the person is younger than 70 or not. As mentioned earlier, this can be thought of as a threshold. Multivariate analysis handles the relationship of multiple features at the same time. A good visualization of this is through the correlation matrix, which is a heatmap that visualizes a linear relationship across all selected features after preprocessing and cleaning the train set.

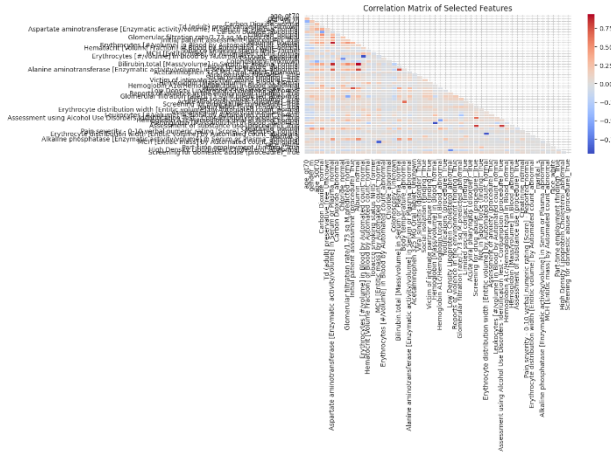


Fig. 7: Multivariate Analysis – Correlation Matrix

The reddish gradient represents a positive correlation to lung cancer, while the blue represents a negative correlation. There are multiple features that show a faint positive correlation to patients associated with lung cancer. Such features include Alkaline phosphatase with Aspartate aminotransferase and Albumin and Glomerular filtration. The negative correlated features, on the other hand, include MCH Entitic Mass and race of white individuals.

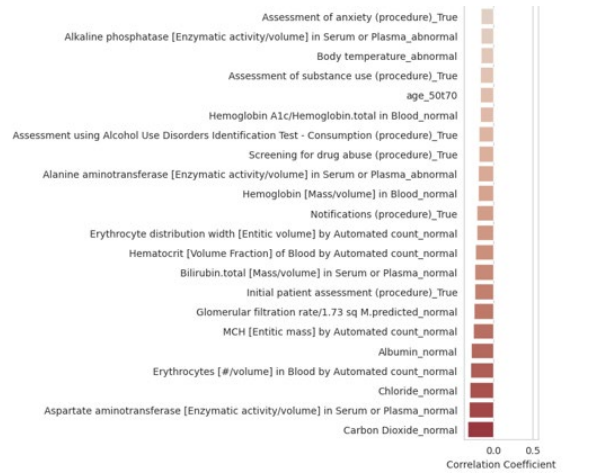


Fig. 8a: Multivariate Analysis – Feature Correlation (Lower Bound)

Another visualization for multivariate analysis is the feature correlation, and it shows how each feature correlated to lung cancer. The reddish gradient shows how each feature has a negative correlation to cancer. Such features include normal carbon Dioxide levels, volume of aspartate aminotransferase, and normal chloride levels. This can imply normal levels of these said features has high negative correlation with lung cancer.

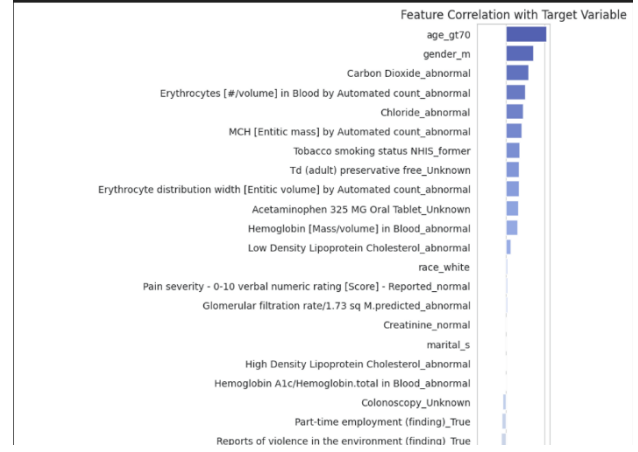


Fig. 8b: Multivariate Analysis – Feature Correlation (Upper Bound)

For the upper part, the feature correlation shows that age_gt70 column is positively correlated to cancer as was discussed earlier with bivariate analysis. Next to the age is gender_m and abnormal carbon dioxide levels as the next two high positively correlated features. It can also be said that abnormal levels of such features are highly correlated with lung cancer. A potential outlier in the dataset is also the reason why that feature is removed in the feature selection. The scc feature or the squamous cell carcinoma is a subtype of lung cancer. The feature may be associated with lung cancer, but it is not clear indicator that anyone with lung cancer can have it. This feature can cause the model to bias towards a subset of cancer types. The group discovered that the scc is not measured unless someone is suspected of having cancer. This mistake can cause potential data leakage. Because of the model's bias, if scc is not measured or observed in the dataset, the model will perform poorly. Analysis highlights the relationship of a single feature to the frequency of observations it has. The coding process basically involves using countplot, while the unique values that are used for the graph were derived through one-hot encoding due to their previously categorical nature. They were then scaled through standardization. In this case, the standardized values become numerical representations. Thus, these two decimal labels would serve if a person were older than 70 or not. Based on the graph, it is apparent that there are a great number of patients who are younger than 70.

D. Initial Model Training

a. Logistic Regression

The initial training of the model, a considered set of steps were followed to accommodate the nature of the algorithm and characteristics of the dataset. In this case, since Logistic Regression is sensitive to feature scaling due to the gradient-based optimization behind it, we utilized a StandardScaler to normalize the input features. This scaling process ensures that all features contribute equally to the model's learning process and avoids disproportionate influence from higher magnitude features. The data pipeline was constructed using Pipeline from scikit-learn, chaining the standardization process and the SGDClassifier model set to use log_loss (logistic loss) for optimization, with class_weight='balanced' to mitigate the effects of class imbalance in the data. The use of

class_weight='balanced' automatically adjusts the weights inversely proportional to class frequencies, which helps the model treat both majority and minority classes with appropriate emphasis during training. The model's random state was fixed (random_state=42) to ensure reproducibility of results across different runs.

Stratified Cross-Validation	
Fold 1	0.85714286
Fold 2	0.85558583
Fold 3	0.88210818
Fold 4	0.8499336
Fold 5	0.85051546
Average F1 score	0.8591

Fig. 6: Stratified Cross-Validation Results for Logistic Regression

On Fig. 5, After establishing the pipeline, a 5-fold stratified cross-validation was performed using StratifiedKFold, which ensures that each fold maintains the same class proportion as the entire dataset. As with XGBoost, the evaluation metric chosen for performance was the F1-score, due to its ability to balance precision and recall, particularly useful in imbalanced classification problems such as medical diagnosis. These values suggest reasonably strong performance across all folds with a fairly consistent trend. The average F1-score calculated was approximately 0.8591, indicating a solid predictive performance. While the variation between the folds was slightly more pronounced than in the XGBoost case, the range still demonstrates acceptable model stability. This performance serves as a robust starting point for later stages of optimization or ensemble modeling, particularly in scenarios requiring balanced classification such as early cancer detection.

b. Support Vector Machine

The model's preprocessing pipeline was designed to account for the specific requirements of the algorithm. Support Vector Machines are highly sensitive to the scale of the input features, as the algorithm computes distances between data points in high-dimensional space to find the optimal separating hyperplane. Consequently, we employed StandardScaler to normalize the feature set prior to training. This step ensures that each feature contributes equally to the margin calculation, which is crucial for the convergence and performance of SVC. To streamline this process, a Pipeline object was created that applies both standardization and the SVC classifier in sequence. The model was also initialized with a fixed random_state=42 to guarantee reproducibility of results.

Stratified Cross-Validation	
Fold 1	0.89595376
Fold 2	0.88921283
Fold 3	0.90190337

Fold 4	0.88888889
Fold 5	0.90322581
Average F1 score	0.8958

Fig. 7: Stratified Cross-Validation Results for SVM

The potential class imbalance and the importance of robust evaluation, we applied stratified 5-fold cross-validation using StratifiedKFold, which preserves the class distribution across all training and validation folds. To assess performance, the F1-score was selected as the primary metric, balancing precision and recalling relevant choice in medical and risk-prediction tasks where false positives and false negatives carry significant consequences. These results demonstrate a remarkably stable and consistently high performance across all partitions of the data.

This low variance in fold performance highlights the stability of the SVM model on the given dataset, indicating that its classification performance is not overly sensitive to specific training splits. Such reliability is especially valuable in domains like medical diagnosis, where ensuring consistent detection across subgroups is critical. With its excellent balance of precision and recall and robustness across validations, the SVM model offers a compelling baseline for further development and hyperparameter tuning in predictive pipelines.

c. Extreme Gradient Boosting

To report the initial training of the Extreme Gradient Boosting (XGBoost) model, a group of particular preparatory steps were implemented to achieve the best performance in case of the peculiarities of the algorithm and the data. Our first step was the following: we used the deliberately chosen unscaled X_train_for_trees which was preprocessed earlier. Such a so-called parameter is essential since such tree-based modeling methods (as XGBoost) are not scale-sensitive to features. Their decision-making process is dependent on determining the best split points according to the value levels not on the intensity of the features and the feature scaling process would be irrelevant in this model. Prior to training, there was one last technical cleanup performed: the names of all the features were sanitized by having all special characters like brackets removed since it is a precautionary step to preclude possible issues encountered with the XGBoost library during: the serialization of models, or their visualization.

Another important feature in the arrangement of the baseline model was to be able to take into consideration the innate problem of the class imbalance with the data. The training data had 4,993 and 1,794 cases of the majority and minority classes, respectively (label 0, no cancer and label 1, cancer). To avoid the model becoming biased towards the majority class, we used the scale_pos_weight parameter in the XGBClassifier. The score of this parameter was determined as a quotient of negative class instances counts and positive class instances counts, which was approximately 2.78. We can make the model give an extra penalty to classifying a real cancer case as a non-cancer case by assigning

scale_pos_weight a value of 2.78 so that it will push hard to classify cancer cases and non-cancer cases correctly by reducing the weight of positive (cancer) class, effectively imposing a penalty of roughly 3 times on incorrectly classifying a true cancer case. This compels the model to give much consideration to the minority class, which is of the essence, in a scenario of medical diagnosis, where failing to identify a cancer case (a false negative) are critical. The model was further set up to use random_state=42 as this makes sure that the outcomes obtained by the model are completely reproducible.

Cross-Fold Validation	
Fold 1	0.89010989
Fold 2	0.89615932
Fold 3	0.89674682
Fold 4	0.9010043
Fold 5	0.89044944
Average F1 score	0.8949

Fig. 8: Cross-Fold Validation Results for XGBoost

Having gathered and cleaned the data, and having adjusted the model to minimize the impact of the class imbalance, the original XGBoost-based model has been trained over the resulting subset of 52 features. We got a solid estimate of its performance by 5-fold cross-validation the training data immediately, and as the measure of performance to be counted to provide an overview of the performance of imbalanced data sets, we use the F1-score. The cross-validation resulted in a very satisfying and almost homogenous choice of F1-scores in Fig. 8. The average F1-score that was obtained was 0.8949, which was very high with a staggeringly low standard deviation of only 0.0041. This small value of a variance indicates high stability of a model, that is, its performance is not highly sensitive to modification in the training data and a high possibility of generalizing. This excellent stable performance provides a good criterion of high quality based on which the results of further hyperparameter tuning will be evaluated.

E. Error Analysis

a. Logistic Regression

The model achieved a respectable accuracy of 91.75%. A deeper look into the confusion matrix reveals important aspects of its predictive behavior. The model correctly identified 1142 True Negatives (no cancer) and 415 True Positives (has cancer). However, it also made two types of errors: 107 False Positives (Type I Errors) and 33 False Negatives (Type II Errors). False Positives, where the model incorrectly predicts cancer, can lead to patient anxiety, unnecessary follow-up procedures, and increased healthcare costs. More critically, the 33 False Negatives, where the model failed to detect actual cancer cases, pose a significant risk, as they can result in delayed diagnosis, progression of the disease, and potentially worse patient outcomes. The precision of 79.50% indicates that roughly 20.5% of the positive predictions were incorrect, directly reflecting the False

Positives. Conversely, a recall of 92.63% means the model successfully identified most actual cancer cases, but the 7.37% it missed represents the critical False Negatives. The F1 score of 85.57% offers a balanced view of precision and recall. In a medical context, the severity of False Negatives often outweighs that of False Positives, making efforts to further reduce these missed diagnoses paramount, even if it means a slight increase in False Positives.

Fig.

b. Support Vector Model

The SVC achieved an accuracy of 0.9576 and an F1-score of 0.9174. However, it also scored a 0.8929 recall score. Based on these values, the Support Vector Classifier did astoundingly well. According to the confusion matrix, Class 0 cases or True Negatives were predicted at 1225, while Class 1 or True Positives were scored at 400. This means that the model has accurately guessed that someone has lung cancer or not. These high numbers are the reasons why the SVC scored such high score. However, it also scored some mistakes such as the False Positives and Negatives. The matrix shows that there are 24 false positives or a prediction mistake that someone has cancer and the opposite is false positive at 24 while false negatives are at 48.

In these cases, it is important to consider the recall score because they are the ones that help determine, medically, if someone has cancer or not. The recall score is determine by the amount of false negatives the model has predicted. It is important to reduce this value as much as possible.

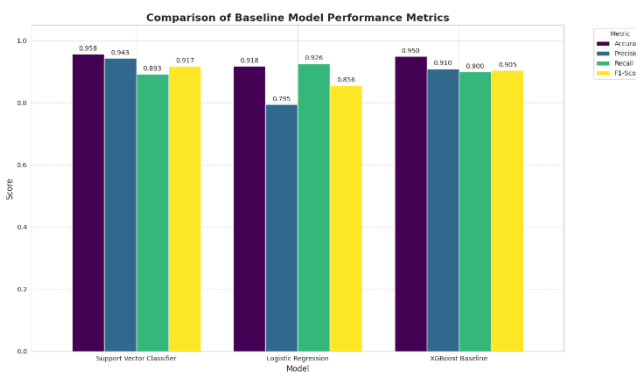
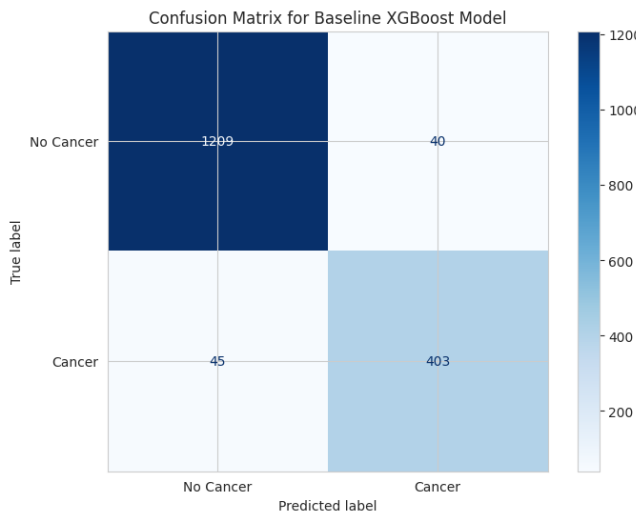
c. Extreme Gradient Boosting

After its successful training and validation, the trained XGBoost baseline model was tested on the unseen test set to carry out an evaluation of the model in the real-world. The model showed good predictive characteristics as the overall accuracy was 95% and weighted average F1-score was 0.95. An in-depth analysis of classification report shows a fine-grained output of two classes. On the majority class ("No Cancer") the model scored very well, with a precision of 0.96 and a recall of 0.97. It means that the model is very effective in the correct recognition of non-lung cancer patients, which reduces the amount of false positives in this patient category. The model performed well and attained a high value on the minority class ("Cancer"), precision, and recall, which were 0.91 and 0.90, respectively. It indicates that the parameters of the scale_pos_weight are helpful since the baseline model was capable of finding the patterns of the minority class without being too biased against the majority.

Further examination of the confusion matrix gives a detailed perspective of the predictive behavior of the model and the errors that were made in what type of manner. It is true that, the model provided results of 1,209 of 1,249 non-cancer patients with True Negatives (TN), reflective of its high specificity. It also was able to predict 403 of the 448 patients with cancer as True Positives (TP) and thus has high sensitivity. Nevertheless, the discussion of misclassifications is the most important. False Positives (FP), generated by the model are 40, and this is the type I error. Medically, these refer to false positives in health patients who would be falsely

identified as suspects of cancer and subjected to unnecessary panic and further tests. More importantly, this model produced 45 False Negatives (FN) or Type II error. Such are the most life-threatening mistakes in medical diagnostic setting since they reflect the real cancer patients who have been misdiagnosed as healthy individuals and thus their treatment may only be provided late.

These types of errors are directly reflected in the metrics contained in classification report. A recall of 0.90 of the cancer class represents that the model managed to capture 90 percent of all the true cancer cases within the test set but could not detect the other 10 percent (the 45 false negatives). The coordinate 0.91 shows that the model was correct in predicting that a patient had cancer 91 percent of the times, and the remaining 9 percent of correct predictions were false alarms. Although the baseline XGBoost model is obviously strong and relatively balanced, the fact that there are 45 instances when the model fails to predict the presence of cancer is reminiscent of the main difficulty and the most important issue to be improved. The final objective of a clinical screening tool is to limit these false negatives, so the improvement of recall is the key aim of the following stage of hyperparameter tuning.



F. Improving Model Performance

a. Logistic Regression

The model already demonstrated strong performance, a deeper look at its misclassifications revealed an area of concern typical in clinical prediction contexts: 33 false negatives, representing actual cancer cases the model failed to

detect. In high-stakes domains such as medical diagnostics, minimizing such errors is paramount. Thus, during the model improvement phase, the central goal was not solely to maximize the F1-score, but more critically to improve recall for the “cancer” class, thereby reducing the number of undetected positive cases.

A GridSearchCV approach was employed for hyperparameter tuning. Despite being computationally more exhaustive than randomized search, Grid Search provided a complete and structured evaluation of all parameter combinations. This level of rigor ensured optimal parameter selection, which is crucial when dealing with imbalanced medical data. The search space (param_grid_LogReg) was carefully designed to test various combinations of regularization penalties ('l1', 'l2', 'elasticnet'), regularization strengths (alpha values of 0.0001 to 0.01), and l1_ratio (relevant for elastic net). All models were evaluated using 5-fold cross-validation, with the evaluation metric set to accuracy for consistency with conventional classification benchmarking. The best-performing model was found using the configuration: 'alpha': 0.01, 'penalty': 'l1', 'loss': 'log_loss', and 'l1_ratio': 0.15.

This setup yielded promising results on the test set, achieving an accuracy of 93%, with class-specific performance metrics reflecting meaningful improvements: a recall of 92% and F1-score of 88% for the cancer class (label 1). Importantly, the model successfully identified 415 out of 448 cancer cases. While 33 cancer cases were still missed (false negatives), this marked an improvement over the baseline and signified progress toward the goal of enhanced sensitivity. The precision of 85% showed a modest compromise, indicating some increase in false positives, a tradeoff considered acceptable given the medical context. This tuning phase reflects the delicate balance between improving recall and maintaining overall performance. The use of class balancing via class_weight='balanced' and a robust grid search helped refine the model toward clinical reliability, where failing to detect a true cancer case (false negative) could lead to dangerous delays in treatment. The updated model thus represents a step forward in developing a safer, more responsive diagnostic tool within the constraints of linear modeling.

b. Support Vector Model

The model’s performance initially did an outstanding and consistent job performing its predictions in its pipeline model with metrics that score above 90%. The model excelled in a various instances distinguishing on the two classes on whether or not a person has cancer. To improve this, the SVM model utilizes a GridSearchCV approach which incorporates a 5-fold validation.

After evaluating 48 parameter configurations across 240 fits, the GridSearchCV found the most optimal and balanced set of parameters which include {'C': 100, 'gamma': 0.001, 'kernel':, 'rbf'}. Afterwards, the model uses the X_test_final and the y_test for fitting and prediction, the model has scored an improved accuracy of 0.9519, a recall 0.8839, and an F1-score of 0.9167. Comparing these scores, the tuned model

fared better than its pipeline and baseline model. However, the recall score, which is highly regarded, is lower.

c. Extreme Gradient Boosting

The model results, the error analysis revealed 45 false negatives, i.e. real positive cases the model failed to detect. The greatest priority should be recognizing the need to minimize such errors in a clinical setting. That is why, during the model improvement phase, the increase of the overall F1-score was not the primary task but rather the boost of the model recall of the "Cancer" class in particular so that the number of false negatives, which can be dangerous, would be decreased. To accomplish that, we have used RandomizedSearchCV that was more computationally efficient compared to a full GridSearchCV. The approach has the advantage of letting one sample the hyperparameter space in a random manner, and a certain trade off between the randomness that it allows and the guarantee that one can get by utilizing a totally exhaustive search, and it is quite efficient to find good hyperparameter combinations without the computationally prohibitive cost involved in testing all combinations.

A parameter distribution (param_dist_refined) was custom-built to seek a better exploration of recall and precision tradeoff and avoid the overfitting menace. Most importantly, we used the scoring metric of the search as a recall and provided specific instructions to the algorithm to find the combination of hyperparameters that would maximize the potential of the model in finding any true positive cases. This trade-off was managed by key parameters use in the search space as the way to tame the model. The scale_pos_weight values were tested on [1, 2.78, 3] to check neutral weighting, the computation of balance in the baseline and an even stronger weighting in favour of the minority category. Regularization parameters such as shallow tree depths (max_depth of 3, 4 or 5), stochastic sampling of data (subsample) and features (colsample_bytree) and L1 (reg_alpha) and L2 (reg_lambda) regularization were examined in order to achieve robustness and to avoid overfitting.

The RandomizedSearchCV was set up to complete 550 iterations with 3-folds cross-validation, and the result is 1,650 model fits. The results of this comprehensive search were finding a new, better set of hyperparameters. Best combination was: {'subsample': 0.9, scale_pos_weight': 3, reg_lambda': 1.0, reg_alpha': 0.1, max_depth': 4, learning_rate': 0.1, colsample_bytree': 0.8}. The choice of scale_pos_weight: 3 (the most aggressive setting) is rather important as well since it proves that the further enhancement of the minority class was the definite solution of enhancing the recall. The rest of the parameters, a set moderate max_depth of 4 and subsample of 0.9, shows that the optimum results were captured when this aggressive approach on recall strategies was coupled with powerful regularization methods to ensure that the model can continue to generalize well. The best cross-validation score of 0.9169 was reported by the search having the mean recall as the scoring parameter of our model. This positive tuning process delivered another, optimized XGBoost model, which is prepared to be evaluated

on the test set one last time that will numerically assess whether it is better than the baseline.

III. DISCUSSION

A. Model Performance Summary

a. Logistic Regression

The tuning of hyperparameters of the logistic regression model led to recalibrating significantly its predictive behavior. The F1-score of the altered model was 0.880, which posed an improvement of 2.423 over its base model. This improvement was largely attributed to the high rise in precision which grown to be 0.847 as compared to the previous 0.795. This shows that the regularization activity that was part of tuning process created more confident and accurate predictions of the positive class. In exchange of this, a small drop in the recall of 0.926 to 0.915 was observed as a trade-off.

Visual confirmation of this shift is the confusion matrix of the tuned Logistic Regression model. False Positives (FP) (the number of healthy patients falsely flagged) shrank by 31 to only 74 in the tuned version when compared to the baseline model (not shown). This however saw a trade-off in the number of False Negatives (FN) as the number increased to 38 cells as compared to 33. Although the F1-score on the whole was increased and shows a more balanced performance, this trade-off reveals one of the prime issues of this model consistency, and now it will also give fewer false alarms in exchange of being missed a little more of the actual cases.

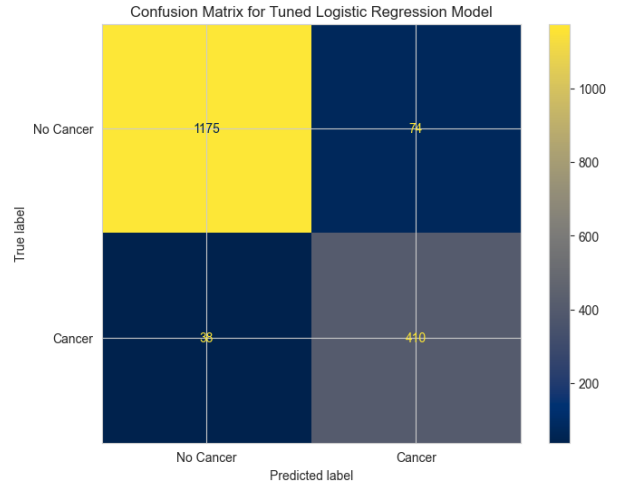


Fig. 9: Confusion Matrix for the tuned Logistic Regression Model

b. Support Vector Model

The tuning of the hyperparameter performed on the Support Vector Machine (SVM) model resulted in a much more refined decision boundary of the model, but with a rather insignificant effect on the general performance data. The F1-score of the cancer class that was learnt using tuned SVM was 0.917 and was practically the same as the baseline performance. This implies that the default SVM model was already close to operating at its optimal level at this data set and there were small changes in it, instead of a significant one made due to tuning.

This regarded as a small shift is observable in the analysis of the confusion matrix of the tuned SVM. The model had a correct identification of 1,220 (True Negatives) non-cancer cases and 404 (True Positives) cancer cases. On the one hand, False Positives were much fewer (40 in the baseline compared to 29 in the tested), which means that the precision improved a bit. Although, this came with an increment in false Negatives of 45 to 44 which translated to insignificant change in recall. On the last note, SVM model, in its baseline and tuned forms was found as a high accuracy instrument being capable of confirming cancer cases but not the most suitable model in terms of maximizing all the possible cases in hand as conveyed by its comparatively low recall when compared with other models.

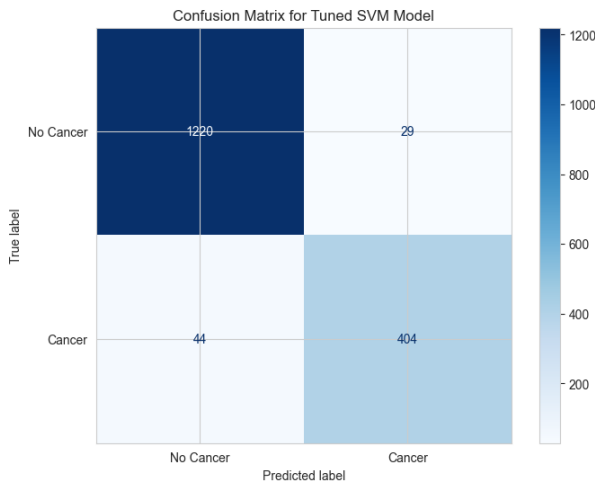


Fig. 10: Confusion Matrix for the tuned SVM Model

c. Extreme Gradient Boosting

The hyperparameter tuning of the XGBoost model that was specifically maximized to the recall perform had the greatest clinical impact. The fitted model had a recall of cancer 0.931 which is much higher than the baseline 0.900. This 3.1 percent enhancement in recall is directly convertible to a significant fall in the cases of missed instances of cancer. A substantial improvement of 0.22 in the recall was also healthy since the F1-score improved to 0.916 which was an indicator that the number of recalled data was not getting sacrificed drastically on the help of precision that only decreased a notch to 0.901. Such a good result proves the validity of the given tuning strategy and especially the choice of augmenting the `scale_pos_weight` parameter, which compelled the model to focus more on the identification of the minority class.

This success can be seen clearly through the confusion matrix of the tuned XGBoost model. False Negatives (FN) also decreased by 31 as compared to the baseline model. This part of the outcome is the most significant of the whole model improvement process since that corresponds to 14 additional cancer patients that have not been missed by the model. Although this also resulted in a marginal rise in False Positives (FP) of 40 to 46, it is a very positive trade as it is used in a medical screening scenario. False positive (a healthy patient who loses a chance to be tested) cost is not worth the benefit of diagnosing a patient with a cancer. The fine-tuned XGBoost performer may, thus, be the most reliable safety net that can be effectively tuned to the main goal of maximizing the number of the actual detected cancer cases.

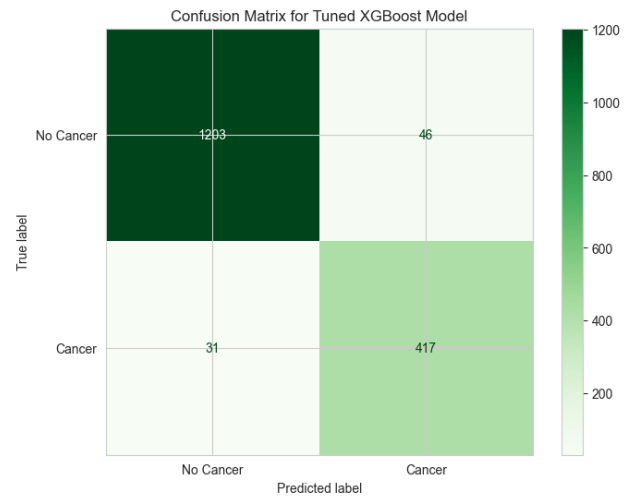


Fig. 11: Confusion Matrix for the tuned XGBoost Model

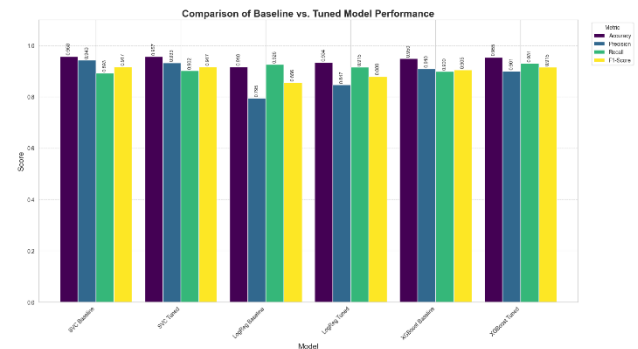


Fig.12: Comparison of Baseline vs. Tuned Model Performance

d. Overall Model Comparison

Comprehensive comparison within all six models associated with three baseline and three tuned models can help identify the peculiarities of performance and emphasize the high influence of the targeted hyperparameter tuning. The results of the final performance summary chart vividly demonstrate that the Support Vector Machine (SVM) started with the best baseline F1-score (0.917) but it was almost unchanged as the result of tuning. On the contrary, the Logistic Regression model and XGBoost model performed significantly better, and their F1-scores increased by 2.4 and 1.1 percent, respectively. It means that these models were more optimizable, which could be achieved because of tuning. The recall score was recorded as the highest among the linear models, implying that the tuned Logistic Regression model is a balanced model in terms of precision (0.847) and recall (0.915).

Nevertheless, when comparing it along the guidelines of clinical usefulness with regard to cancer screening, tuned XGBoost can be undoubtedly considered the best model at hand. Although the highest F1-score was attained by the tuned SVM by a thin margin (0.917 as opposed to 0.916 scored by XGBoost), this metric does not represent the entire clinical context. The ultimate goal in the context of a screening application is to reduce false negatives, and the tuned XGBoost model did it excellently. It also has the ability to name those patients who really have cancer which is shown by achieving the highest recall among the models of 0.931.

This can be directly mirrored in its confusion matrix that displays the fewest number of missed cancer cases (31) than that of any model. The tuned SVM missed 44 cases, although its F1-score was high, as compared to 38 by the tuned Logistic Regression.

IV. CONCLUSION

A. Key Insights

In the course of this project a definite pattern was observed in what concerns the trade-off between various machine learning models, as well as how powerful the effects of target optimization can be. Each of the initial baseline models used a different predictive strategy: the Logistic Regression model was a high-recall, low-precision dragnet, capable of accurately detecting high proportion of cancer instances at the expense of significant number of false positives. On the other hand, the Support Vector Machine (SVM) represented a scalpel of high precision, low recall, which was certain of the positive, but more cautious, and erroneous. The initial balance was better in case of the baseline XGBoost model, thanks to the fact that the XGBoost method inherent `scale_pos_weight` parameter is able to deal with the issue of class imbalance.

We can explain this superior performance of ensemble-based XGBoost model compared to the linear Logistic Regression as well as the kernel-based SVM due to the ability of the former to find complex and non-linear interrelationships and hierarchies, existing within the high-dimensional EHR data. The most important phase was the hyperparameter tuning one, which demonstrated that the XGBoost model had the best potential to be improved, clinically relevant. Although tuning SVM resulted in the negligible changes and thus it was already at substantially high performance, optimization of the XGBoost model resulted in the considerable overall improvement of the recall performance when recalled specifically. This shows that in more complicated models, an averagely default setting is not enough; the performance can be improved by a fit the optimization strategy to the actual, real goals of the problem the one here being minimizing the missed cancer diagnosis.

Based on similar research using the same dataset, comparing models such as: XGBoost, Random Forest (RF), SVM, and K-Nearest Neighbors (KNN). All the models compared with the XGBoost model achieve the best results [2]. Proving that the research and our research shows XGBoost is the best suited model in achieving risk prediction lung cancer.

B. Conclusion Summary

In this paper, a strong machine learning framework was designed and confirmed to predict the occurrence of lung cancer early in a simulated patient population. By a careful application of data cleaning, context-specific imputation, methodic feature reduction and parameter optimization, we showed that it was possible to develop an extremely useful clinical screening measure. The soundness and real-world utility of our results were secured by our methodology that involved proactive removal of features subject to target leakage (such as the `sec` biomarker) and a more stringent policy of taxonomy and preprocessing (we performed this step in the split-then-preprocess protocol). The optimized XGBoost model (the last one) became the best solution, with

the remarkable recall of 0.931 on the cancer category of the unseen test set. This is literally an equivalent of accurate prediction of all the actual cancer instances (out of 448) in a data set by the model with only 31 instances out of 448 being False Negative (which would be very serious and critical), showing a significant improvement and advantage over any other models.

The approach proves that the Synthea dataset is a high-fidelity and useful tool towards creating and validating clinical prediction models with the ability to maintain the privacy of data in a privacy-preserving way. Based on the outcomes, the terminal can see that the data has a rich, complex signal, which suggests that it can be further used with sophisticated algorithms such as XGBoost. This is a good proof-of-concept that similar data-centric and objective-oriented machine learning strategies would attain a similar success when applied to live EHR data, and ultimately lead to better patient outcomes due to earlier and more precise detection of the disease.

C. Recommendations

Though this project has managed to create a high-recall model, there are other ways in which it can be improved in future. During the data preprocessing, more advanced imputation strategy, which computes the missing values using the information in similar patient profiles, like applying a KNNImputer, might give more accurate data than the current median-based imputation strategy. Moreover, we might want to go one step further and not only one hot encode numbers corresponding to laboratory results, but also bin the values into clinically significant categories (e.g. values corresponding to low level, normal value, high value) and then one hot encode the bin. This can be done because the real numbers corresponding to a laboratory result may not reflect a linear relationship with the output. In feature engineering, it may be useful to come up with explicit interaction terms (e.g. multiplying age and a smoking variable) or sum scores (e.g. a total comorbidity count) to compress more meaningful predictive signals into the available data.

Theoretically, in terms of modelling development, more complex algorithms are an apparent follow-up. The application of a deep learning model, e.g. Tabular Neural Network, might unveil complex feature interplay that is otherwise not possible to learn with the help of tree based models. Alternatively, the performance improvement might be achieved by testing other state-of-the-art gradient boosting frameworks such as LightGBM or CatBoost, since they each present their own set of unique efficiencies and strategies of working with data. Lastly, hyperparameter tuning process can be further optimized. With a less sample-intensive algorithm such as Bayesian Optimization there would be a way of smartly traversing the parameter space to converge on a good solution more quickly. This would then be followed up by additional fine-grained GridSearchCV on the small promising portion of the parameter space discovered by the coarse-grained one so as to make the specialisation of the model further.

V. ACKNOWLEDGMENT

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